Report on Udemy Course: Learn 3D Printing – Fused Deposition Modeling (FDM)

1. Introduction (About the Course)

1. Name of the Course:
Learn 3D Printing – Fused Deposition Modeling (FDM)

2. Platform:
Udemy

3. Instructor / Institute Name:
Danish Technological Institute

4. Course Duration:
Approximately **31 minutes**

This report presents a summary and analysis of the online certification course *Learn 3D Printing – Fused Deposition Modeling (FDM)* completed through Udemy. The course provides a 360-degree introduction to FDM, one of the most widely used additive manufacturing technologies. It covers the fundamentals of the FDM process, materials, design considerations, printing process, post-processing, quality assurance, and health and safety precautions.

The course is designed to provide foundational knowledge for students and engineering learners interested in modern manufacturing technologies and rapid prototyping.

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2. Objectives of the Course

The primary objectives of this course were:

- To understand the **basic principle of 3D printing using FDM technology**.
- To learn how material is deposited layer by layer to create 3D objects.
- To study the **materials used in FDM printing**.
- To understand **design guidelines and limitations** of FDM-manufactured parts.
- To learn about **print orientation, infill, and lattice structures**.
- To understand **post-processing methods** such as sanding and finishing.
- To gain awareness of **health and safety risks** associated with 3D printing.
- To understand quality assurance methods including **CT scanning and inspection**.

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3. Topics / Modules Covered

The following modules were covered in the course:

1. 2. Introduction to 3D Printing and FDM
3. 4. Pros and Cons of FDM
5. 6. Design Possibilities
7. 8. Design Guidelines
9. Materials for FDM
10. Applications of FDM
11. Health and Safety
12. The Printing Process

    Post Processing

    Quality Assurance

    References and Additional Resources

    Quizzes and Assessments

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4. Key Learnings

The course helped develop a strong understanding of **Fused Deposition Modeling (FDM)**.

What is FDM?

FDM stands for **Fused Deposition Modeling**. It is an additive manufacturing process in which thermoplastic material in filament form is melted and extruded through a nozzle. The material is deposited layer by layer according to the CAD model until the final component is formed.

Material Form

The raw material used in FDM is generally in the form of **thermoplastic filament** wound on a spool.

Important Design Properties

- Layer-by-layer manufacturing
- Anisotropic strength behavior
- Surface roughness due to layer lines
- Need for support structures for overhangs
- Infill density affects strength and weight

Benefits of Infill / Lattice Structures

- Reduced material usage
- Lower weight
- Reduced print time
- Sufficient internal strength
- Better structural optimization

Advantages of FDM

- Low cost
- Easy to use
- Suitable for prototypes
- Widely available materials
- Good for complex geometries

Limitations of FDM

- Lower surface finish compared to other AM processes
- Visible layer lines
- Lower dimensional accuracy
- Anisotropic mechanical properties
- Longer time for large parts

Safety Learnings

- Use proper ventilation
- Wear gloves while handling hot parts
- Avoid touching hot nozzle and bed
- Use protective glasses if required
- Ensure safe electrical connections

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5. Quizzes (Questions and Answers)

The course included multiple quizzes. Some important questions and answers are listed below:

Quiz 1

Q1. What is FDM an abbreviation for?
Answer: Fused Deposition Modeling

Q2. What is the shape of FDM raw materials?
Answer: Filament / wire spool form

Q3. What are the benefits of using infill structures?
Answer: Reduced material consumption, reduced weight, and optimized strength

Q4. Which advantage characterizes FDM?
Answer: Low-cost and rapid prototyping capability

Quiz 2

Q5. Why are FDM printed parts often sanded?
Answer: To improve surface finish and remove visible layer lines

Q6. What is the purpose of CT scanning in FDM parts?
Answer: Internal defect inspection and quality assurance

Q7. Which health and safety risk may occur in FDM?
Answer: Burns from hot nozzle / fumes from heated material

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6. Skills Developed

Through this course, the following skills were developed:

- Basic knowledge of **additive manufacturing**
- Understanding of **FDM process workflow**
- Knowledge of **material selection**
- Awareness of **design-for-manufacturing principles**
- Understanding of **quality control and inspection techniques**
- Safety awareness in 3D printing labs
- Practical understanding of rapid prototyping methods

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7. Conclusion

The *Learn 3D Printing – Fused Deposition Modeling (FDM)* course on Udemy provided a clear and practical introduction to one of the most important additive manufacturing processes used in engineering and product development.

The course successfully covered theoretical concepts along with practical design and safety aspects. It is highly beneficial for engineering students, designers, and beginners interested in 3D printing technology.

This course enhanced the understanding of how modern manufacturing technologies are transforming product development, prototyping, and industrial design.

Course Completion Status: 100% Completed

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Quiz : 1

Question 1:

What is FDM an abbreviation for?

- ○ Filament Dosing Machine
- ○ Fast Design Method
- ◉ Fused Deposition Modeling
- ○ Filament Deposition Manufacturing

Question 2:

What is the shape of FDM raw materials

- ◉ Filament
- ○ Pills
- ○ Powder
- ○ Both powder and pills

Question 3:

What is the material deposition movement pattern?

- ○ The nozzle moves in X-Y-direction, and the build plate always moves downwards in Z-direction
- ○ The nozzle moves in X-, Y- and Z-direction, and the build plate is always fixed
- ○ Both nozzle and build plate rotate, nozzle moves upwards and build plate downwards in z-direction
- ◉ The relative movement between nozzle and build plate may vary, and depends on printer brand

Question 4:

Which advantages characterize FDM?

- ○ There are many different materials in different colors to choose from for FDM
- ○ There are many FDM printers to choose among – at all price levels from a few hundred Euro
- ○ FDM is great for 3D-print beginners, and for prototyping
- ◉ All of the above

Question 5:

Which limitation characterize FDM?

- ○ FDM printed parts usually require support
- ○ The surface of FDM printed parts has clearly visible layers
- ○ FDM is slow for batch production compared with e. g. SLS
- ◉ All of the above

Question 6:

How is the tensile strength of FDM-printed parts?

- ○ It depends on the material type, in which direction (X, Y or Z) the tensile strength is highest
- ◉ Strongest in X-Y-direction
- ○ Uniform strength in X-Y-Z directions (= good isotropic properties)
- ○ Strongest in Z-direction

Question 7:

What are the benefits of using infill/lattice structures?

- ○ It will save material and weight
- ○ It will print faster and save time
- ○ It will save money
- ◉ All of the above

Question 8:

What are the design properties for FDM parts?

- ○ Maximum bridging length without support is 25 mm
- ◉ Quality of holes depends on the orientation, and vertical holes have highest quality — Correct ✓
- ○ Minimum printed hole diameter is 5 mm
- ○ Expected linear tolerance is +/- 0.1 %

Question 9:

What is true about materials for FDM?

- ◉ PLA and ABS are among the most common FDM materials — Correct ✓
- ○ PP and PE are used for high performance applications
- ○ Opposed to SLS printing, PA cannot be used for FDM printing
- ○ FDM printed parts are always printed with pure plastic materials

Quiz: 2

Question 1:

Which of the below may be a health and safety risk using FDM?

- ○ Print head vibrations due to fast X-Y movements?
- ◉ **Filament particle inhalation?** Correct ✓
- ○ FDM printer noise level?
- ○ Filament etching risk?

Question 2:

Which protective equipment and precautions are recommended for FDM print?

- ○ Safety shoes
- ○ Safety glasses
- ◉ **Good ventilation / fresh air supply** Correct ✓
- ○ Work gloves

Question 3:

What may be the purpose of ensuring right orientation of a part for FDM printing?

- ○ To minimize support structure
- ○ To minimize build time
- ○ To optimize part strength
- ◉ **All of the above** Correct ✓

Question 4:

Why are FDM printed parts often sanded?

- ○ To remove marks after support
- ○ To smoothen visible layers
- ○ To remove residues after gluing/assembling parts
- ◉ **All of the above** Correct ✓

Question 5:

What is needed for re-using filament material?

- ○ A boiler, a mixing rod and an extruder
- ○ Solvent, an oven and an extruder
- ◉ **A shredder and an extruder** Correct ✓
- ○ A compactor, glue and an extruder

Question 6:

What may be the purpose of CT scanning an FDM printed part?

- ○ To check dimensions
- ○ To check for internal defects
- ○ To create a 3D file for print (“reverse engineering”)
- ◉ **All of the above** Correct ✓

Great job! You are ready to move on to the next lecture.

You got 9 out of 9 correct.

What you know

- What is true about materials for FDM?
- What are the design properties for FDM parts?
- What are the benefits of using infill/lattice structures?
- How is the tensile strength of FDM-printed parts?
- Which limitation characterize FDM?
- Which advantages characterize FDM?
- What is the material deposition movement pattern?
- What is the shape of FDM raw materials
- What is FDM an abbreviation for?

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Great job! You are ready to move on to the next lecture.

You got 6 out of 6 correct.

What you know

- What may be the purpose of CT scanning an FDM printed part?
- What is needed for re-using filament material?
- Why are FDM printed parts often sanded?
- What may be the purpose of ensuring right orientation of a part for FDM printing?
- Which protective equipment and precautions are recommended for FDM print?
- Which of the below may be a health and safety risk using FDM?

Retry quiz | Continue

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